

Pyrolizer Heating Jacket

My role is to simply provide a certain amount of energy to the surface of the wall. Everything is simulated as if the heating jacket is concurrent pipes.

```
> restart;  
  
> print("====Key Equations====");  
eq1 := Q__req = m*Cp*(T__hot1 - T__hot2); #Energy Balance from  
page 478 fundametals of heat and mass transfer  
eq2 := Q__req = h__j*A*DeltaT__lm;          #Page 483 fundametals  
of heat and mass transfer  
DeltaT__lm := ((T__hot1 - T__cold)-(T__hot2 - T__cold))/(ln(  
(T__hot1-T__cold)/(T__hot2-T__cold))); #Page 482 fundametals of  
heat and mass transfer  
R__gas := 8.314:
```

"====Key Equations===="

$$eq1 := Q_{req} = m C_p (T_{hot1} - T_{hot2})$$

$$eq2 := Q_{req} = h_j A \Delta T_{lm}$$

$$\Delta T_{lm} := \frac{T_{hot1} - T_{hot2}}{\ln \left(\frac{T_{hot1} - T_{cold}}{T_{hot2} - T_{cold}} \right)} \quad (1.1)$$

```
> #1 is in  
#2 is out  
#Hot is oil/salt/water  
#Cold is the wall
```

```
print("====Specifications====");  
T__hot1 := 605.5797078 + 273.15; #Kelvin (Defined in Furnace  
Specs) (Assumes no loss of energy in pipes)  
m := 0.7609; #kg/s (From Furnace)
```

```

D_pyrolizer := 0.318; #m
D_jacket := 0.348; #m
L_1 := 2.05; #m # heating zone
L_2 := 2.57; #m # constant temp zone
Q_1 := 23550; #W # Required energy for pyrolizer heating zone
Q_2 := 20070; #W # Energy needed to maintain temperature

T_cold_in := 280 + 273.15; #K, wall at biomass inlet
T_cold_out := 550 + 273.15; #K, wall at end of L1 / all of L2

print ("====Properties of ternary eutectic mixture====");
rho := 2050; #kg/m3
Cp := 2300; #J/kg/K
k_fluid := 0.55; #W/m/K
F_mu_salt := T_K -> 0.0852*exp((3.51e4)/(R_gas*T_K))/1000;

```

"====Specifications===="

$$T_{hot1} := 878.7297078$$

$$m := 0.7609$$

$$D_{pyrolizer} := 0.318$$

$$D_{jacket} := 0.348$$

$$L_1 := 2.05$$

$$L_2 := 2.57$$

$$Q_1 := 23550$$

$$Q_2 := 20070$$

$$T_{cold_in} := 553.15$$

$$T_{cold_out} := 823.15$$

"====Properties of ternary eutectic mixture===="

$$\rho := 2050$$

$$Cp := 2300$$

$$k_{fluid} := 0.55$$

$$F_{\mu_{salt}} := T_K \mapsto \frac{0.0852 \cdot e^{\frac{35100.}{R_{gas} \cdot T_K}}}{1000} \quad (1.2)$$

```
> print ("====Assume temperature change is constant====");

T_hot_mid := T_hot1 - Q_1/(m*Cp); #salt temp leaving L1
T_hot2 := T_hot_mid - Q_2/(m*Cp); #salt temp leaving L2
T_hot_avg1 := (T_hot1 + T_hot_mid)/2; # average salt temp in
L1
T_hot_avg2 := (T_hot_mid + T_hot2)/2; # average salt temp in
L2
```

"====Assume temperature change is constant===="

$$T_{hot_mid} := 865.2731032$$

$$T_{hot2} := 853.8049905$$

$$T_{hot_avg1} := 872.0014055$$

$$T_{hot_avg2} := 859.5390468$$

(1.3)

```
> # Hydraulic diameter
```

$$D_h := 4 \cdot A_c / P_w;$$

$$v := m / (\rho \cdot N_ch \cdot A_c); \quad \# \text{ velocity per channel}$$

$$D_h := \frac{4 A_c}{P_w}$$

$$v := \frac{0.0003711707317}{N_{ch} A_c}$$

(1.4)

```
> # ===== Spiral Baffle Channel Calculations =====
```

$$\text{Pitch} := 0.07; \quad \# \text{ m}$$

$$t_baffle := 0.005; \quad \# \text{ m}$$

$$w_ch := \text{Pitch} - t_baffle; \quad \# \text{ m}$$

$$\text{GAP} := (D_jacket - D_pyrolizer)/2; \quad \# \text{ m}$$

Flow Area and Wetted Perimeter

$$A_c := w_ch \cdot \text{GAP};$$

$$P_w := 2 \cdot w_ch + 2 \cdot \text{GAP};$$

Hydraulic diameter & Velocity

$$D_h := 4 \cdot A_c / P_w;$$

$$v := m / (\rho \cdot A_c);$$

```
# Flow Path Lengths
```

```
L__path1 := L__1 *(Pi * D__pyrolizer/Pitch);
```

```
L__path2 := L__2 *(Pi * D__pyrolizer/Pitch);
```

```
Pitch := 0.07
```

```
t_baffle := 0.005
```

```
w_ch := 0.065
```

```
GAP := 0.0150000000
```

```
Ac := 0.0009750000000
```

```
P_w := 0.1600000000
```

```
D_h := 0.02437500000
```

```
v := 0.3806879299
```

```
L_path1 := 29.25720359
```

```
L_path2 := 36.67854303
```

(1.5)

```
> #Size CHECK
```

```
#cooling, Page 491, fundametals of heat and mass transfer. Uses  
Sieder-Tate correlation
```

```
# For Section 1
```

```
Re__1 := rho*v*D__h/F_mu__salt(T__hot_avg1);
```

```
Pr__1 := Cp*F_mu__salt(T__hot_avg1)/k__fluid;
```

```
Gz__1 := Re__1*Pr__1*D__h/L__path1;
```

```
Nu__1 := 1.86*Gz__1^(1/3) * (F_mu__salt(T__hot_avg1)/F_mu__salt  
(T__cold_out))^0.14;
```

```
h__1 := Nu__1*k__fluid/D__h; #(Book uses U)
```

```
# For Section 2
```

```
Re__2 := rho*v*D__h/F_mu__salt(T__hot_avg2);
```

```
Pr__2 := Cp*F_mu__salt(T__hot_avg2)/k__fluid;
```

```
Gz__2 := Re__2*Pr__2*D__h/L__path2;
```

```
Nu__2 := 1.86*Gz__2^(1/3) * (F_mu__salt(T__hot_avg2)/F_mu__salt  
(T__cold_out))^0.14;
```

```
h__2 := Nu__2*k__fluid/D__h; #(Book uses U)
```

```
Re_l := 1762.752682
```

```
Pr_l := 45.12750835
```

```
Gz_l := 66.27420854
```

```
Nu_l := 7.230326303
```

$$\begin{aligned}
 h_1 &:= 163.1458243 \\
 \Re_2 &:= 1643.257016 \\
 Pr_2 &:= 48.40912625 \\
 Gz_2 &:= 52.86464104 \\
 N_2 &:= 6.771727932 \\
 h_2 &:= 152.7979636
 \end{aligned}$$

(1.6)

```
> # Heat transfer area
A__enhanced_per_m := 3.14 * D__pyrolizer;
```

```
A__av1 := A__enhanced_per_m * L1;
```

```
A__av2 := A__enhanced_per_m * L2;
```

$$A_{enhanced_per_m} := 0.99852$$

$$A_{av1} := 0.99852 L1$$

$$A_{av2} := 0.99852 L2$$

(1.7)

```
> # Section 1: co-current, wall heated up
dT1__in := T__hot1 - T__cold_in;
dT1__out := T__hot_mid - T__cold_out;
LMTD__1 := (dT1__in - dT1__out)/ln(dT1__in/dT1__out);
```

```
# Section 2: wall constant
```

```
dT2__in := T__hot_mid - T__cold_out;
```

```
dT2__out := T__hot2 - T__cold_out;
```

```
LMTD__2 := (dT2__in - dT2__out)/ln(dT2__in/dT2__out);
```

```
# Area without baffle fins
```

```
A__bare1 := Pi * D__pyrolizer * L__1;
```

```
A__bare2 := Pi * D__pyrolizer * L__2;
```

```
#Area of the baffle fin
```

```
A__baffle1 := 2 * GAP * L__path1;
```

```
A__baffle2 := 2 * GAP * L__path2;
```

```
# Required Area vs Available Area
```

```
A__av1 := A__bare1 + A__baffle1;
```

```
A__req1 := Q__1/(h__1 * LMTD__1);
```

```

A__av2 := A__bare2 + A__baffle2;
A__req2 := Q__2/(h__2 * LMTD__2);

# Check if design is valid
if A__av1 > A__req1 and A__av2 > A__req2 then
    print("Design Successful");
else
    print("Design Failed");
end if;

```

$$dT1_{in} := 325.5797078$$

$$dT1_{out} := 42.1231032$$

$$LMTD_1 := 138.6088444$$

$$dT2_{in} := 42.1231032$$

$$dT2_{out} := 30.6549905$$

$$LMTD_2 := 36.08584219$$

$$A_{bare1} := 2.048004251$$

$$A_{bare2} := 2.567498012$$

$$A_{baffle1} := 0.8777161076$$

$$A_{baffle2} := 1.100356291$$

$$A_{av1} := 2.925720359$$

$$A_{req1} := 1.041415426$$

$$A_{av2} := 3.667854303$$

$$A_{req2} := 3.639929405$$

"Design Successful"

(1.8)

```

> #Assumes no added pressure from coiling up and down sides,
gravity pressure is negligible.
# Laminar friction factor for straight duct
f__1 := 16/Re__1; # Fanning friction factor, laminar
DP__1 := 4*f__1*(L__path1/D__h)*(rho*v^2/2);

f__2 := 16/Re__2;
DP__2 := 4*f__2*(L__path2/D__h)*(rho*v^2/2);

```

```
DP__total := DP__1 + DP__2;
```

```
 $f_1 := 0.009076712896$ 
```

```
 $DP_1 := 6473.495632$ 
```

```
 $f_2 := 0.009736760498$ 
```

```
 $DP_2 := 8705.706282$ 
```

```
 $DP_{total} := 15179.20191$ 
```

(1.9)

```
> #Summary of design, Bellow script generated using CLAUDE OPUS 4.6
```

```
print("====Summary====");
printf("Section 1 (Heating Zone, L1 = %.2f m):\n", L__1);
printf("  Re = %.4f\n", Re__1);
printf("  h  = %.2f W/m2K\n", h__1);
printf("  T_salt_in  = %.2f K (%.2f C)\n", T__hot1, T__hot1-273.15);
printf("  T_salt_out = %.2f K (%.2f C)\n", T__hot_mid, T__hot_mid-273.15);
printf("  T_wall_in  = %.2f K (%.2f C)\n", T__cold_in, T__cold_in-273.15);
printf("  T_wall_out = %.2f K (%.2f C)\n", T__cold_out, T__cold_out-273.15);
printf("  A_req = %.4f m2, A_av = %.4f m2\n", A__req1, A__av1);
printf("\nSection 2 (Constant Temp Zone, L2 = %.2f m):\n", L__2);
printf("  Re = %.4f\n", Re__2);
printf("  h  = %.2f W/m2K\n", h__2);
printf("  T_salt_in  = %.2f K (%.2f C)\n", T__hot_mid, T__hot_mid-273.15);
printf("  T_salt_out = %.2f K (%.2f C)\n", T__hot2, T__hot2-273.15);
printf("  T_wall      = %.2f K (%.2f C)\n", T__cold_out, T__cold_out-273.15);
printf("  A_req = %.4f m2, A_av = %.4f m2\n", A__req2, A__av2);
printf("\nFlow:\n");
printf("  v = %.4f m/s\n", v);
printf("  DP_total = %.2f kPa\n", DP__total/1000);
print("====");
```

```
"====Summary===="
```

```
Section 1 (Heating Zone, L1 = 2.05 m):
```

```
Re = 1762.7527
```

```
h  = 163.15 W/m2K
```

T_salt_in = 878.73 K (605.58 C)
T_salt_out = 865.27 K (592.12 C)
T_wall_in = 553.15 K (280.00 C)
T_wall_out = 823.15 K (550.00 C)
A_req = 1.0414 m2, A_av = 2.9257 m2

Section 2 (Constant Temp Zone, L2 = 2.57 m):

Re = 1643.2570
h = 152.80 W/m2K
T_salt_in = 865.27 K (592.12 C)
T_salt_out = 853.80 K (580.65 C)
T_wall = 823.15 K (550.00 C)
A_req = 3.6399 m2, A_av = 3.6679 m2

Flow:

v = 0.3807 m/s
DP_total = 15.18 kPa

"=====

(1.10)